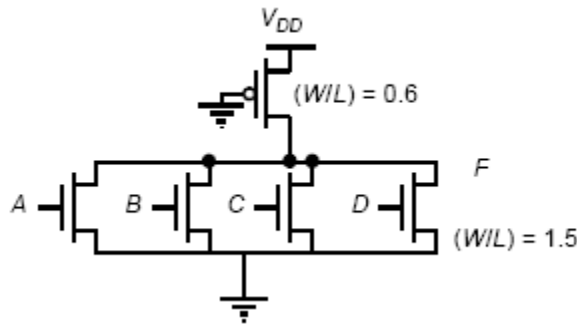


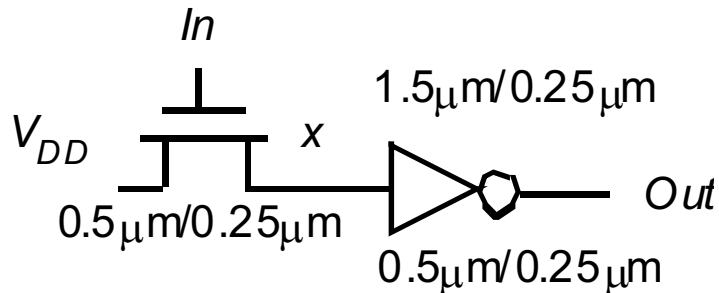
Homework 2

(Due 10th October November at the beginning of the class)

- (1) Assume that the inputs to an N-input XOR gate are uncorrelated and uniformly distributed; derive the expression for the switching activity factor.
- (2) Consider the following circuit



- (a) What is the output voltage if only one input is high? If all four inputs are high?
- (b) Compare your analytically obtained results to a SPICE/Cadence Spectre simulation.
- (3) Consider the following pass transistor logic circuit discussed in class:



First determine the voltage at nodes X and Out when the input is logic “1”. After this step, please design the level restorer circuit. Show the voltage level at the node X when input is logic “1” with the restorer. Size the restorer suitably so that you can reliably transmit logic “0”. You need to do this using Cadence simulation, no hand calculation will be accepted

- (4) Now, convert the above circuit to a transmission gate logic. Demonstrate that you have full swing at node X .